MITHIL BATRA

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EDUCATION

National Institute of Technology, Delhi — **2023 – Present**
B.Tech. in Computer Science and Engineering — CGPA: 8.93

Stepping Stones Sr. Sec. School, Chandigarh — **2022 – 2023**
CBSE Board (XII) — Percentage: 91.8%

Seth Anandram Jaipuria School, Vasundhara — **2020 – 2021**
CBSE Board (X) — Percentage: 94.6%

SKILLS

Languages	Java, Python, C, C++, JavaScript, TypeScript
Frameworks & Libraries	Spring Boot, Spring, Next.js, React, Flask, OpenCV, Tailwind CSS
Backend & APIs	REST APIs, Microservices, API Integration, Webhooks, Node.js
Databases	PostgreSQL, MySQL, MongoDB
Cloud & DevOps	GCP, AWS, Azure, Docker, Git, GitHub
AI/ML	Machine Learning, Deep Learning, scikit-learn, Pandas, Artificial Intelligence
Core	Data Structures & Algorithms, OOP, Design Patterns, Software Engineering
Tools	Postman, Arduino, Figma

WORK EXPERIENCE

HSBC — Software Engineering Intern — **June 2026 – July 2026**

- Enhanced an internal **TDS Chatbot** by improving intent recognition, response generation, and workflow automation for employee tax queries.
- Integrated the **Vidura microservice** into the GPS portal through service integration, API integration, and environment configuration.
- Supported the **Panorama** application using **Spring Boot** and **GCP** through production bug fixes and code quality improvements.
- Worked on application debugging, service integration, environment configuration, and production deployment workflows.

PROJECTS

Pit Wall – F1 Podium Prediction & Analytics Dashboard | *Python, Flask, scikit-learn, Pandas, FastF1, Docker, REST APIs*

- Built a full-stack **machine learning application** to predict Formula 1 podium finishes using live FastF1 timing and telemetry data.
- Engineered a leakage-safe **feature engineering pipeline** with time-ordered splits and walk-forward backtesting across three seasons.
- Trained and benchmarked **Random Forest and gradient-boosting models** using Top-3 Hit Rate and PR-AUC evaluation metrics.
- Automated live race-data ingestion and daily model retraining, and containerized the application with **Docker** for cloud deployment.

GitHub | Live Demo

Resolvr – Incident Management and Alerting App | *Next.js, Tailwind CSS, PostgreSQL, Node.js, REST APIs*

- Built a full-stack **incident management platform** using Next.js, Node.js, PostgreSQL, and REST APIs.
- Implemented a Kanban-based incident workflow supporting manual and **webhook/API-driven** incident creation.
- Designed on-call scheduling and escalation policies with an automated **multi-channel incident notification** system.
- Integrated incident timelines and runbooks to centralize incident context and streamline resolution workflows.

GitHub

AI-Powered Automated Mock Interview Platform | *Next.js, Tailwind CSS, PostgreSQL, Google Gemini API*

- Built a full-stack **AI interview platform** using Next.js, PostgreSQL, and Google Gemini API for automated interview simulation.
- Implemented role-, industry-, and experience-based **question generation** with AI-powered answer evaluation and personalized feedback.
- Designed a PostgreSQL backend for storing interview sessions, generated questions, responses, and evaluation results.
- Integrated Gemini API workflows to automate question generation and interview feedback, reducing manual mentor involvement.

GitHub | Live Demo

CERTIFICATIONS

- **Cassandra (Udemy)** – Gained expertise in distributed database management, fault tolerance, and horizontal scalability. Applied knowledge to optimize real-world data-intensive applications, ensuring high availability and reliability.
- **Azure Services (Microsoft)** – Acquired practical experience in designing, deploying, and managing scalable cloud solutions. Strengthened ability to integrate services like Azure Functions, Storage, and DevOps pipelines for enterprise use cases.
- **C++ Proficient Practitioner (Edchart)** – Strong understanding of advanced C++ programming concepts, with proficiency in STL, memory management, and optimization for developing efficient and reliable software..

ACHIEVEMENTS & LEADERSHIP

- Deputy General Secretary – Google Developer Student Clubs (GDSC), NIT Delhi
- Executive – Training and Placement Cell, NIT Delhi (Calling and Social Media Team)
- Quarter-Finalist – Flipkart Grid | L'Oréal Sustainability Challenge
- Won Gold at MST Tournament 2025 held at NIT Jaipur
- Secured **99.467 percentile** in JEE Main, among the top 0.5% of candidates nationwide
- Qualified JEE Advanced with an **All India Rank of 5376**
- Solved **800+** problems across coding platforms

COURSEWORK

Software Engineering, Data Mining, Database Management Systems, Quantum Computing, Machine Learning, Artificial Intelligence, Natural Language Processing, Computer Vision, Pattern Recognition, Probability and Statistics, Optimization Techniques